10 General Science

Sound Investigation

Name:_____ Teacher: _____ Due: Monday 1/8/11

Mark: /35

Your task is to investigate how a sound interacts with its environment.

1. How does a microphone convert sound waves into an electrical signal? Draw a diagram to help you explain. (4 marks)

2. How does a speaker convert an electrical signal into a sound wave? Draw a diagram to help you explain. (4 marks)

3. Compare how sound travels through solids, liquids and gas and give their comparative speeds. (6 marks)

4. What does a change in wave frequency do to the sound? (2 marks)

5. What does a change in wave amplitude do to the sound? (2 marks)

6. Using diagrams explain how sonar works. (5 marks)

7. Using a flow diagram explain how Miss Cerny, using her super-human hearing, hears someone whispering in the back row of a classroom. Starting from the

sound wave and finishing in Miss Cerny’s brain. (Include the parts of the ear) (12 marks)